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WEEK 2 SUBMISSION
REQUIREMENTS ENGINEERING

1. Introduction

This document presents the Requirements Engineering deliverable for the Moderated Learning Management System (LMS) specified in the CS4433 Software Engineering project. The LMS is intended for use within a secondary school environment and supports the activities of Administrators, Lecturers, Heads of Department (HoDs), and Students. A fundamental business rule of the system is that all marks must undergo moderation by the relevant HoD before being released to students. In addition, every mark entry, modification, moderation decision, and recalculation must be recorded in a secure audit trail.

The project follows the Waterfall Software Development Life Cycle (SDLC), while verification and validation activities are structured using the V-Model. This document establishes the requirements baseline from which the design, implementation, testing, and deployment phases will be derived. The planned implementation is a menu-driven C++17 application using SQLite as the persistence layer and designed to operate effectively in an offline school environment.

1.1 Purpose and scope

The system manages users, departments, courses, lecturer allocations, student enrolments, assignments, submissions, resubmissions, extensions, marking, feedback, moderation, mark release, submission lifecycle states, and audit records of all mark-related changes.

1.2 Definitions and abbreviations

Term	Definition
LMS	Learning Management System.
HoD	Head of Department; reviews and approves marks for courses in the department.
FR	Functional requirement: a behaviour or service the system must provide.
NFR	Non-functional requirement: a measurable quality or constraint.
MoSCoW	Priority classification: Must, Should, Could, Won't have in this release.
Submission	A student's uploaded or recorded assignment work and its submission timestamp/status.
Moderation	The HoD's review and decision on marks before marks are released.
Audit trail	A record of a mark change including who changed it, when, old/new mark, and reason.
Effective Deadline	The assignment deadline or a revised deadline granted through an approved extension.

2. Stakeholder analysis

Stakeholders were identified from the case study. Their influence, principal concerns, and involvement are summarised below. The four user roles in the application are Administrator, Lecturer, HoD, and Student. The Principal and IT Officer are important external stakeholders whose policies and constraints influence acceptance.

Stakeholder	Main interests / needs	Influence	System involvement
Principal	Simple system; marks remain hidden until the HoD has checked them.	High	Approves operating policy and accepts the release process.
Administrator	Reliable management of users, departments, courses, lecturer assignments and enrolments.	High	Uses administration functions and maintains accurate records.
Lecturer	Create assignments and deadlines, grant individual extensions, mark work, and give feedback; would prefer prompt release.	High	Creates assessment work and marks student submissions.
Head of Department (HoD)	See marks before release, moderate marks, apply consistent late penalties, and control release.	High	Reviews/moderates marks and releases approved results.
Student	See assignments and deadlines, submit/resubmit before the deadline, and receive released marks with written feedback.	Medium	Views course work, submits assignments, and views released results.

IT Officer	Offline operation, protected passwords, acceptable performance for 800 students, maintainability and backup.	High	Supports installation, security, performance and data protection.

2.1 Stakeholder engagement

Method	Participants	Purpose / output
Structured interview or requirements review	Principal and HoD	Confirm approval authority, mark visibility, late penalties and release policy.
Walk-through of marking workflow	Lecturers and HoD	Confirm assignment setup, extensions, marking, moderation and feedback needs.
Scenario-based review	Students or student representatives	Check submission, resubmission, late-work and feedback expectations.
Technical review	IT Officer	Validate offline operation, security, backup, performance and deployment constraints.
Requirements sign-off	All stakeholder representatives / supervisor	Resolve outstanding conflicts and baseline the SRS before detailed design.

3. Conflict and ambiguity resolution

ID	Statement or source	Problem	Resolution
C1	LE: marks as soon as marked. PR and HoD: nothing visible before the HoD checks.	Direct conflict on when students see marks	Moderation comes first. Students see only Released marks. The lecturer sees their own marks at once, and the HoD releases a whole assignment in one step so release is quick.
C2	LE: wants marks quickly. SR: resubmit before the deadline.	Marking early would lock the student out of resubmitting	A submission can be marked only after its effective deadline has passed.
C3	HoD: “10% per day for up to three days”	Undefined: what a day is, 10% of what, and the boundaries	A day is a started 24-hour period after the effective deadline. Exactly at the deadline is on time, and exactly 24 hours late is day 1. The penalty is 10% of the **maximum mark** per day, deducted from the raw mark, with a floor of zero.
C4	SR: late work not rejected. HoD: zero after three days.	“Receive zero” could mean rejection	Late work is always accepted and recorded. After day 3 the system gives zero, and the lecturer can still write feedback.
C5	LE: extensions. HoD: penalty rule.	Does an extension bypass the penalty? Where does the penalty clock start?	An extension changes only that student’s effective deadline, and the penalty is counted from it.

			A reason is mandatory, and the HoD sees extensions during moderation.
C6	SR: “fix a mistake and resubmit before the deadline”	How many times? What happens to old versions? What about after the deadline?	Unlimited resubmission until the effective deadline while the work is unmarked. The latest version counts and earlier versions are kept. No resubmission after the deadline.
C7	HoD: “moderate”	Can the HoD change marks, or only approve?	The HoD can approve, change a mark (reason mandatory) or return it to the lecturer. Every change is audited.
C8	Brief: “all changes to marks recorded”	Is the first entry a “change”? What about penalty recalculation?	Yes to both. Every entry, change and recalculation is logged.
C9	PR: “must be simple”. Others want more features.	Feature wishes against simplicity	MoSCoW prioritisation, a short console menu per role, and Could items only after all Must items work.
C10	IT: offline and unreliable network. Brief: “single computer or local network”.	A shared database file over a network would be unreliable	v1.0 is a single computer with a local database. Network sharing is not supported.
C11	IT: “respond quickly”	Not measurable	2 seconds for login and assignment list, 3 seconds for other operations, on defined hardware with 800 students stored.
C12	IT: “passwords protected”	How?	Salted Argon2id hashes, a unique salt per user, and lockout after repeated failures.
C13	No statement from the administrator	Missing needs	The Administrator role is retained as a separate system role. Where administrator-specific requirements are unavailable from stakeholder interviews, requirements are derived from the case study and reviewed with the supervisor. The Administrator may access audit logs but shall not access student marks (principle of least privilege).

C14	SR: written feedback	When does the student see it?	Feedback is released together with the mark, never earlier.
C15	Not stated	Cut-off for late submissions	Submissions are accepted until the HoD releases marks for that assignment, then the assignment is closed.

Worked Example: Late Penalty Calculation

An assignment is marked out of 50. A student receives a raw mark of 40 and submits 30 hours after the effective deadline.

According to the agreed policy, each started 24-hour period counts as one late day.

30 hours late = 2 late days

Penalty = 10% × 50 × 2 = 10 marks

Final Mark = 40 − 10 = 30

If the submission occurs after more than 72 hours (three late days), the final mark becomes 0 according to the late-submission policy.

3.1 Assumptions and policy decisions

- The system is a local, menu-driven C++17 application with an SQLite database and must work without internet access.
- The four in-system roles are Administrator, Lecturer, HoD and Student. The Principal and IT Officer are stakeholders, but the case study does not define them as application roles.
- Only the HoD may approve and release marks for courses in their department.
- A lecturer may mark only submissions for courses assigned to that lecturer.
- Students may access only their own submissions, released marks and feedback.
- Late penalty calculation is based on the assignment's available maximum mark, not the student's awarded mark. A late day is defined as any started 24-hour period after the effective deadline, unless revised during stakeholder review.
- An extension is student-specific and changes that student's effective deadline; it does not change the deadline for other students.
- The system records a mark change with actor, timestamp, old mark, new mark and reason. A mark cannot be silently overwritten without an audit record.
- The system clock of the host computer is used for submission timestamps, deadline validation, and late-penalty calculations. Correct system time is therefore an operational assumption.
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4. Software Requirements Specification (SRS)

4.1 Requirement conventions

Each requirement has a unique identifier. 'Must' means essential for the assessed release, 'Should' means important but not essential to the minimum release, 'Could' means desirable if time permits, and 'Won't' means

explicitly excluded from version 1.0. Requirements use observable outcomes so that they can be verified through inspection, demonstration, or test.

4.2 Functional requirements

ID	Area	Priority	Requirement statement	Acceptance / verification criterion
FR-01	Authentication	Must	The system shall allow a registered user to sign in using a unique username and password.	Given valid credentials, the user is authenticated and the correct role-specific menu is displayed.
FR-02	Role-based access	Must	The system shall restrict each operation to users whose role is authorised to perform it.	For each role, an access-control test shall show permitted actions succeed and prohibited actions are denied.
FR-03	User administration	Must	The Administrator shall be able to create, view, update, deactivate and reactivate user accounts.	A test account can be created and updated; a deactivated account cannot sign in.
FR-04	Department and course management	Must	The Administrator shall be able to create and maintain departments and courses and associate each course with one department.	A saved course has a valid department reference and appears under that department.
FR-05	Lecturer assignment and enrolment	Must	The Administrator shall be able to assign lecturers to courses and enrol students in courses.	The resulting lecturer/course and student/course relationships are stored and can be retrieved.
FR-06	Assignment creation	Must	A Lecturer shall be able to create an assignment for an assigned course with a title, description, deadline and maximum mark greater than zero.	An assignment with valid required fields is saved; invalid or missing required values are rejected.
FR-07	Assignment and deadline viewing	Must	A Student shall be able to view assignments and applicable deadlines for courses in which the student is enrolled.	The list contains only assignments for enrolled courses and displays the effective deadline, including an approved individual extension.
FR-08	Submission and resubmission	Must	A Student shall be able to submit assignment work and resubmit it before the applicable deadline; each accepted submission shall have a recorded timestamp.	A valid initial submission and a valid pre-deadline resubmission are recorded with distinct timestamps; an attempt after the permitted window follows FR-09.
FR-09	Late submission and penalty	Must	The system shall accept late work for up to three days after the applicable deadline and apply a deduction of 10% of the assignment's maximum mark per late day; work later than three days shall receive zero under the late policy.	Boundary tests at the deadline and at 1, 2, 3 and more than 3 late days shall produce the documented policy outcome. A late day is defined as any started 24-hour period after the effective deadline.
FR-10	Individual extension	Must	A Lecturer shall be able to grant an extension to an individual student for an assignment in a course taught by that lecturer. Each extension shall specify a revised deadline and a mandatory reason, and the revised deadline shall become the student's effective deadline.	An authorised extension is saved with the revised deadline and is visible to that student; unauthorised users cannot create it.
FR-11	Marking and feedback	Must	Lecturer shall be able to assign a mark from zero to the assignment maximum and record written feedback for a submission. Marks and feedback shall remain hidden from students until moderation and release have been completed.	A mark outside the valid range is rejected; a valid mark and feedback are stored against the submission.
FR-12	Mark moderation	Must	The HoD shall be able to review marks for courses in their department and approve a mark, amend a mark with a recorded reason, or return a mark for correction before release.	Students cannot see a mark while it is awaiting moderation; the HoD's decision and timestamp are recorded.
FR-13	Mark release and student viewing	Must	Only marks approved by the HoD shall be released, and students shall be able to view only their own released marks and feedback.	A released approved mark is visible to its student; unapproved marks and other students' results remain inaccessible.
FR-14	Mark-change audit trail	Must	The system shall create an audit record for every mark entry, mark modification, moderation adjustment, and penalty recalculation. Each record shall contain the actor, timestamp, previous value, new value, and reason.	A mark-change test produces an audit record containing all five required fields, and ordinary users cannot edit or delete it.
FR-15	Submission lifecycle	Must	The system shall maintain a valid status for each submission, including submitted, late where applicable, marked, moderated/returned and released states.	State-transition tests reject invalid transitions and allow only transitions defined in the lifecycle rules.
FR-16	Validation and error handling	Must	The system shall validate required fields, numeric ranges, dates and referenced records, and shall report invalid input without terminating unexpectedly.	Invalid input produces a useful error message, leaves data consistent, and returns control to the relevant menu.
FR-17	Record lookup	Should	Authorised users shall be able to find assignments, submissions and results by relevant course and student identifiers.	A search returns matching authorised records and does not expose records outside the user's permissions.
FR-18	Audit Lot Access	Must	The HoD shall be able to view audit records for courses within their department, and the Administrator shall be able to view audit records across the entire system.	Audit records can be filtered by assignment, course, or student and are displayed only to authorised users.
FR-19	Password Management	Should	Any authenticated user shall be able to change their own password after successfully providing their current password.	Password change succeeds only when the current password is correct.
FR-20	Notification of	Could	The system shall notify students at their next login that newly	After mark release, a notification is displayed

	Released Marks		released marks are available.	to the relevant student on login.

4.3 Non-functional requirements

ID	Quality attribute	Priority	Requirement statement	Verification criterion
NFR-01	Performance	Must	On the target school computer, login and opening a student's assignment list shall each complete within 2 seconds for a representative database containing records for 800 students, excluding initial application startup.	Run a repeatable performance test with a documented 800-student dataset; record timings and hardware.
NFR-02	Offline availability	Must	Core functions shall operate without an internet connection after the application and required libraries have been installed.	Disconnect the machine from the network and test login, course/assignment viewing, submission recording, marking and moderation.
NFR-03	Password protection	Must	The system shall not store plaintext passwords; it shall store salted password hashes produced by a recognised password-hashing library.	Inspect database contents and verify authentication against valid and invalid passwords.
NFR-04	Authorisation and confidentiality	Must	The system shall enforce role permissions for every protected operation and prevent a student from viewing another student's private submission, mark or feedback.	Execute role-based negative tests for all roles, including direct attempts to access another user's records.
NFR-05	Data integrity	Must	The database shall use primary keys, foreign keys and suitable constraints to prevent orphaned records and invalid mark ranges; related multi-step changes shall be transactional.	Run database constraint tests and simulate a failed multi-step operation to verify rollback.
NFR-06	Auditability	Must	Every mark change shall preserve the old and new values, actor, timestamp and reason in an audit record that normal application users cannot alter or delete.	Inspect the schema and test the mark-update operation and attempted audit-record modification.
NFR-07	Usability	Should	A first-time user shall be able to complete the role's basic tasks using labelled console menus and on-screen instructions without editing files or SQL directly.	Conduct a short task-based walkthrough for one user from each role and record completion or confusion.
NFR-08	Maintainability and portability	Should	The application shall be written in C++17, built through CMake, and separate business rules from SQLite access through a data-access layer.	A clean build shall succeed using the documented CMake procedure; inspection confirms database queries are contained in the data-access layer.
NFR-09	Reliability and recovery	Should	The installation guide shall document a database backup and restore procedure, and a backup shall be restorable without manually editing database contents.	Follow the documented procedure on a test copy and verify that users, assignments, submissions and audit records are recovered.
NFR-10	Installation	Should	A school technician shall be able to install and run the system on a clean Windows or Linux computer within 30 minutes using only the installation guide.	A timed installation trial is completed successfully following the documented procedure.

4.4 Requirements Deliberately Excluded from Version 1.0

The following requirements have been explicitly excluded from the initial release of the Moderated LMS in order to maintain project scope and align with the case-study constraints:

- Web-based interfaces.
- Mobile applications.
- Online payment processing.
- Video conferencing and virtual classroom functionality.
- Automated grading systems.
- Plagiarism detection services.
- Multi-computer concurrent LAN deployment; Version 1.0 assumes operation on a single local computer.
- Direct student visibility of marks before HoD moderation and release.
- Integration with external learning platforms or cloud services.

5. Use-case model

The diagram shows the four application actors and their principal interactions with the LMS. The diagram is a high-level view; detailed rules and exceptions are specified in the fully dressed use cases that follow. The

Principal and IT Officer are stakeholders but are not shown as actors because the case study does not define direct application use cases for them.

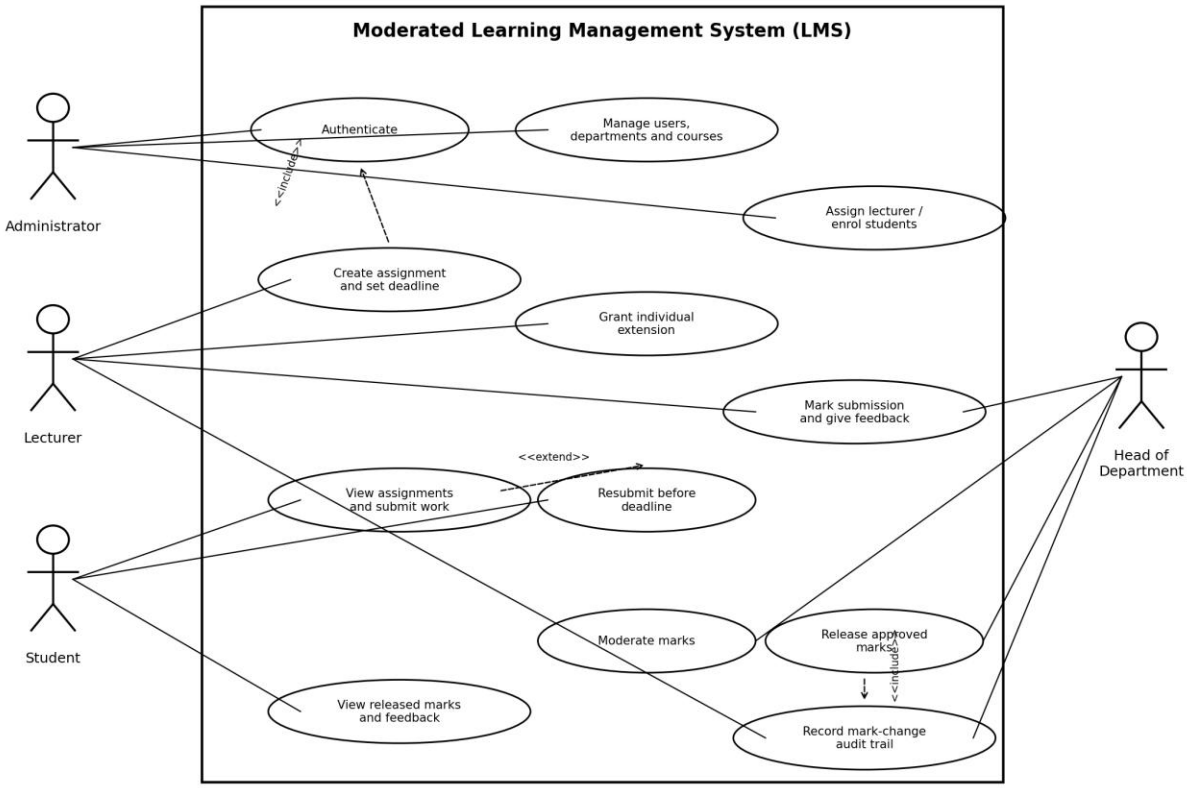


Figure 1: High-level UML-style use-case diagram for the Moderated LMS.

5.1 Actor descriptions

Actor	Description	Main use cases
Administrator	Maintains the institutional records required for LMS operation.	Authenticate; manage users, departments and courses; assign lecturers; enrol students.
Lecturer	Sets assessment work and assesses student submissions.	Authenticate; create assignment; grant individual extension; mark submission; provide feedback.
Head of Department	Ensures marks are reviewed and approved before publication.	Authenticate; review/moderate marks; return marks for correction; release approved marks.
Student	Views course assignments, submits work, and views published results.	Authenticate; view assignments/deadlines; submit/resubmit work; view released marks and feedback.

5.2 Relationships and rationale

- «include» Authenticate: each protected use case requires an authenticated session. This is a common required behaviour rather than an optional extension.
- «include» Record mark-change audit: creating, modifying, moderating, or recalculating a mark must automatically create an audit record as part of the same business operation; it is not an optional user action.
- «extend» Resubmit assignment: resubmission is an optional flow that extends the normal submission use case when a previous submission exists and the applicable deadline has not passed.
- «include» Moderate Marks before Release Marks: a mark cannot be released until moderation has been completed and approved by the HoD.

- The diagram does not force generalisation between the four actors. They have distinct responsibilities and the case study does not require a common specialised actor relationship beyond their shared need to authenticate.
- The final diagram should be checked against the detailed use cases and updated if the lecturer expects every include/extend relationship to be drawn with formal UML arrow directions and labels.

6. Fully dressed use-case specifications

UC-01: Submit Assignment

Scope: Moderated Learning Management System (LMS).

Level: User goal.

Primary actor: Student.

Supporting actors: None directly; the system records the submission.

Stakeholders and interests: Student: wants work recorded and wants to know whether it was submitted on time. Lecturer: needs the correct submission to mark. HoD: needs consistent late-policy treatment and traceable status. IT Officer: needs reliable local storage.

Preconditions: The student is authenticated; the student is enrolled in the course; the assignment exists and is available; the student has permission to submit to it.

Trigger: The student selects an assignment and chooses Submit or Resubmit.

Main success scenario: 1. The system displays the assignment title, instructions, maximum mark and applicable deadline. 2. The student selects the work to submit or enters the submission content supported by the application. 3. The system validates that the assignment and student records exist and that the submission is within the permitted submission window. 4. The system records the submission content/reference and current timestamp. 5. The system determines whether the submission is on time or late. 6. The system sets the submission status to Submitted or Late, as applicable. 7. The system confirms success and displays the recorded submission time and status.

Alternative flow A1: Resubmission before deadline: At step 2, the system detects an earlier submission. If the applicable deadline has not passed, the student confirms resubmission. The system records a new submission version and timestamp, preserves the prior submission history, and identifies the latest valid version for marking. The use case continues at step 5.

Alternative flow A2: Individual extension: At step 1, the system finds an approved extension for this student and assignment. It displays the revised deadline and uses that deadline for lateness checks.

Alternative flow A3: Late submission within the permitted period: At step 3, the system finds that the applicable deadline has passed but the submission is no more than three days late. It accepts and timestamps the submission, marks it as **Late**, and makes it subject to the approved late-submission policy. The lecturer/marking workflow applies the penalty based on the assignment maximum mark. A late day is defined as any started 24-hour period after the effective deadline.

Exception flow E1: More than three days late: If the submission is more than three days late, the system records the submission attempt, applies the agreed late-submission policy, and assigns a final mark of zero. The submission remains available for lecturer feedback and audit purposes.

Exception flow E2: Invalid or unavailable assignment: The system displays a clear error and does not create a valid submission.

Exception flow E3: Storage failure: The system reports that the submission could not be saved, does not display a success confirmation, and leaves the database in a consistent state.

Postconditions on success: A submission record exists with a timestamp and correct status; prior versions are preserved if resubmission occurred; the latest valid submission is identifiable for marking.

Postconditions on failure: No false success is reported and no partially saved submission is treated as valid.

Special requirements: Submission handling must work offline. Input must be validated. Submission history and status must be stored reliably. Students must not be able to view or alter another student's submission.

UC-02: Mark Submission and Provide Feedback

Scope: Moderated Learning Management System (LMS).

Level: User goal.

Primary actor: Lecturer.

Supporting actors: None directly; the system validates and stores the mark and feedback.

Stakeholders and interests: Lecturer: wants to assess work efficiently and give feedback. Student: wants an accurate mark and useful written feedback. HoD: wants marks ready for moderation before release.

Preconditions: The lecturer is authenticated and assigned to the course; the assignment and submission exist; the submission is available for marking.

Trigger: The lecturer selects a submission and chooses Mark.

Main success scenario: 1. The system displays the selected submission, assignment maximum mark and current status. 2. The lecturer reviews the work. 3. The lecturer enters a mark between zero and the maximum mark. 4. The lecturer enters written feedback. 5. The lecturer saves the assessment. 6. The system validates the mark and required references. 7. The system stores the mark and feedback and sets the submission status to Marked/Awaiting Moderation. 8. The system confirms that the assessment is saved but not yet visible to the student.

Alternative flow A1: Late submission: The system calculates late days as started 24-hour periods after the effective deadline and applies the approved late-submission policy.

Alternative flow A2: Lecturer corrects a mark: If a mark already exists, the lecturer enters the corrected mark and provides a reason. The system records the old mark, new mark, actor, timestamp and reason in the audit trail before confirming the update.

Exception flow E1: Invalid mark: If the entered mark is negative or greater than the maximum, the system rejects it, explains the allowed range, and asks the lecturer to correct it.

Exception flow E2: Unauthorised lecturer: If the lecturer is not assigned to the course, the system denies access and does not change the submission.

Postconditions on success: The mark and feedback are stored; status is awaiting HoD moderation; any mark change has a corresponding audit record; the mark is not released to the student.

Postconditions on failure: The stored assessment remains unchanged and no unauthorised mark modification occurs.

Special requirements: Mark saving and audit creation must be atomic. Database queries must be parameterised. The system must enforce role and course assignment checks.

UC-03: Moderate and Release Marks

Scope: Moderated Learning Management System (LMS).

Level: User goal.

Primary actor: Head of Department (HoD).

Supporting actors: Lecturer indirectly, where a mark is returned for correction; Student receives results after release.

Stakeholders and interests: HoD: must check marks before release and apply department policy. Principal: wants results hidden until checking is complete. Lecturer: needs clear correction requests. Student: needs accurate released marks and feedback.

Preconditions: The HoD is authenticated; the course belongs to the HoD's department; the relevant submissions have been marked or are awaiting moderation.

Trigger: The HoD opens a course's marks awaiting moderation.

Main success scenario: 1. The system lists marks awaiting moderation for a course in the HoD's department. 2. The HoD selects a submission and reviews the mark, feedback and applicable late penalty. 3. The HoD approves the mark. 4. The system records the moderation decision, actor and timestamp. 5. The HoD chooses to release the approved marks for the relevant release batch or course. 6. The system verifies that each mark selected for release has been approved. 7. The system changes the approved records to Released. 8. The system confirms release. 9. The student can then view their own released mark and feedback.

Alternative flow A1: Return mark for correction: At step 3, the HoD does not approve the mark and records a reason for return. The system records the decision and sends the item back to the lecturer's correction queue. The mark remains hidden from the student until it is corrected, moderated and released.

Alternative flow A2: Release only approved marks: If a release batch contains both approved and unapproved marks, the system releases only the approved marks or rejects the mixed batch with a clear explanation. It must never publish an unapproved mark.

Exception flow E1: Wrong department: If the HoD attempts to moderate a course outside their department, the system denies the operation and records no moderation decision.

Exception flow E2: No marks awaiting moderation: The system displays an empty-state message and allows the HoD to return to the course menu.

Exception flow E3: Database failure during release: The system reports failure and uses a transaction so the release operation does not leave a partially updated release batch.

Postconditions on success: The moderation decision and timestamp are stored; only approved marks are released; released results become visible only to the relevant students.

Postconditions on failure: Unapproved marks remain hidden; failed release does not leave inconsistent status values.

Special requirements: Release is restricted to the HoD; mark visibility is enforced by the system rather than relying on users to follow instructions; actions must be auditable.

7. Requirements traceability preview

The complete traceability matrix will be consolidated in the final report. The following initial links show how key requirements lead into use cases and future verification artefacts. IDs must be retained consistently in the design, implementation and test stages.

Requirement	Use case(s)	Planned design elements	Planned verification
FR-01, FR-02	All three use cases	Authentication service; User hierarchy; role/permission checks	Authentication tests; role-based negative tests
FR-06, FR-07	UC-01; UC-02	Assignment and Course entities; assignment repository	Assignment validation and visibility tests
FR-08, FR-09, FR-10	UC-01; UC-02	Submission lifecycle; deadline/extension and penalty services	Deadline boundary tests; extension and resubmission tests
FR-11, FR-12, FR-13	UC-02; UC-03	Marking, moderation and release services; submission status	Mark-range, moderation and release tests
FR-14	UC-02; UC-03	MarkAudit entity/table; transactional mark update	Audit completeness and tamper-access tests
NFR-01, NFR-02	UC-01; UC-02; UC-03	SQLite persistence; local deployment	Performance test on 800-student dataset; offline test
NFR-03, NFR-04	All protected use cases	Password hashing and authorisation layer	Password storage inspection; access-control tests

8. Review and acceptance criteria

The requirements baseline is ready for design review when:

- All functional requirements have unique IDs, priorities, and testable acceptance criteria.
- All non-functional requirements specify a measurable target or a repeatable verification method.
- Stakeholder conflicts have a documented resolution or are explicitly flagged for confirmation.
- The use-case model covers the four defined application roles and aligns with the detailed use cases.
- The three use-case specifications define preconditions, trigger, main flow, alternatives, exceptions and postconditions.
- The supervisor/stakeholders confirm unresolved policy points, especially late-day rounding, treatment of submissions more than three days late, and whether one-computer operation is acceptable.
- The agreed SRS baseline is used to derive the design artefacts and V-Model test plan.

9. Items requiring confirmation

The case study does not fully define several operational details. These are deliberately visible rather than silently assumed to be settled policy. They should be raised at the requirements review and then updated in this document if the lecturer or stakeholders give a definitive answer.

- Whether work more than three days late should be retained as a rejected/zero-mark attempt or whether a file may still be stored for audit purposes.
- Whether individual extensions may be modified after a submission has been submitted and recorded.
- Whether resubmission retains all previous file versions or only metadata and the latest version.
- Whether v1.0 is strictly single-user on one computer, or must support multiple users on a local network.
- Whether the school requires a specific password length, reset process, or account lockout policy.
- Whether HoD moderation should permit mark amendments in addition to approval and return-for-correction actions.
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10. Conclusion

This requirements engineering document establishes the initial functional and quality baseline for the Moderated LMS. It identifies the stakeholders, resolves major policy conflicts provisionally, specifies verifiable requirements, and defines the main interactions through a use-case model and three detailed use cases. The next stage is to review and baseline these requirements, resolve the open policy questions, and use the approved SRS to drive the architecture, class design, database design and V-Model verification activities.

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